Module IV

Speed Control of 3 phase Induction Motors

Introduction

- Earlier Induction motors were used in applications requiring a constant speed because variable speed applications have been dominated by DC drives
- Conventional methods for speed control of Induction motors were either expensive or highly inefficient
- Later the availability of thyristors, power transistors, IGBT and GTO have allowed the development of variable speed induction motor drives
- DC motors require frequent maintenance due to the presence of commutators & brushes. Also they cannot be used in explosive & dirty environment
- On the other hand, induction motors particularly squirrel cage are rugged, cheaper, lighter, smaller, more efficient, requires less maintenance and can be operated in dirty & explosive environment .
- Due to these advantages, Three-phase induction motors are the most common machines in industry now & more than 90% of mechanical power used in industry is supplied by 3 phase induction motors.
- Variable speed induction motor drives are expensive than DC drives
- Application include fans, blowers, cranes, conveyors, traction, underground & under water installations etc

Principle of operation

- When a 3 phase supply is given to 3 phase stator winding, a rotating magnetic field is produced in the stator. The speed of rotation of rotating magnetic field is called *synchronous speed(Ns)*

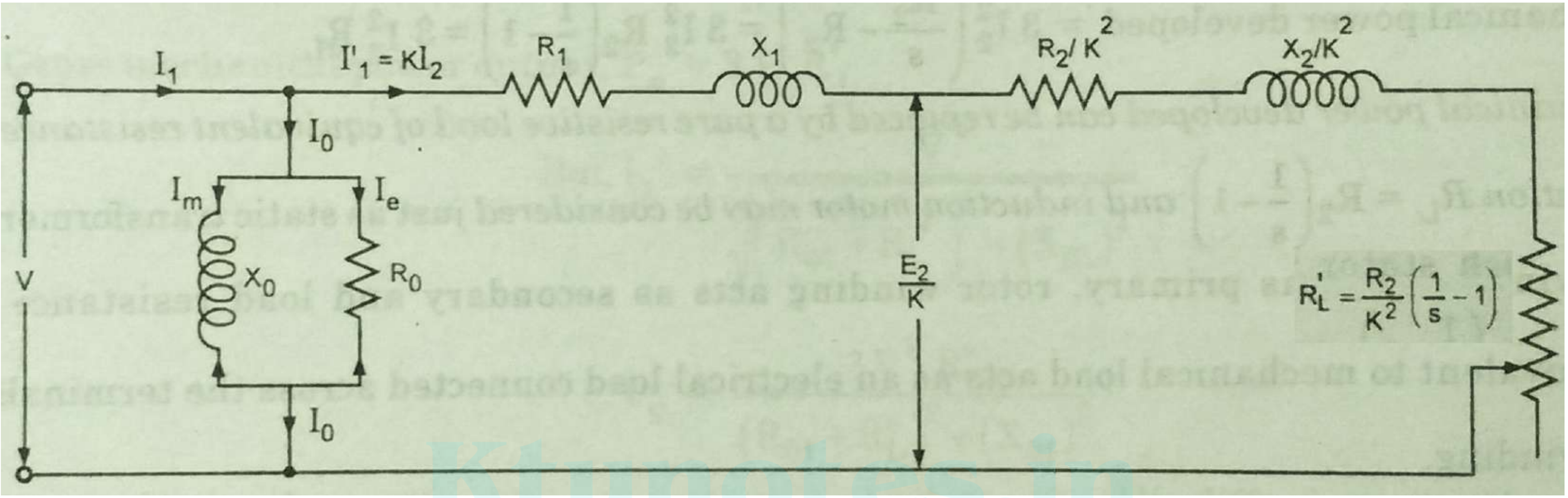
$$N_s = (120f)/P$$

 where, P = number of poles on the stator, f = supply frequency

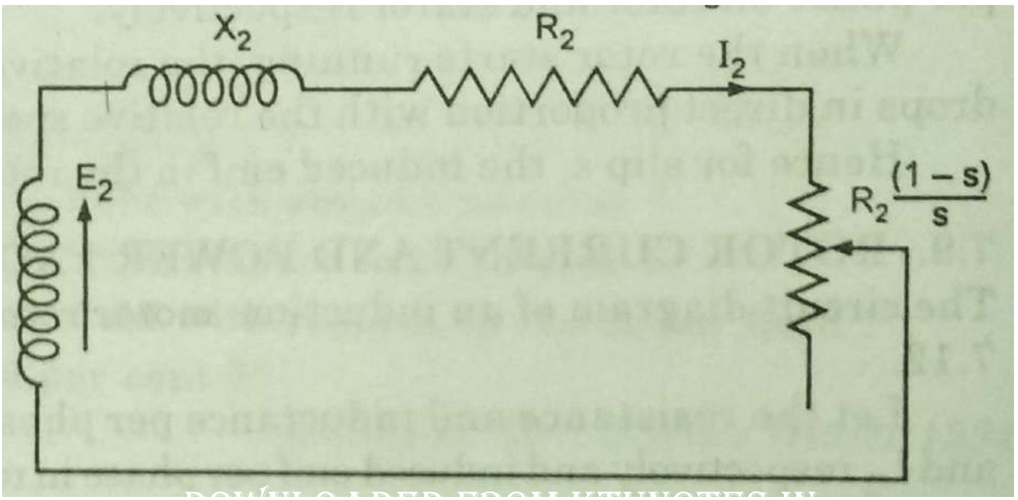
- This rotating magnetic field links the stationary rotor windings and produces an induced emf in the rotor windings
- Since the rotor windings are short circuited for both squirrel cage and wound-rotor, a large current flows through the rotor windings
- Now the situation is exactly like *a current carrying conductor placed in a magnetic field*
- The current carrying conductor placed in a magnetic field experiences a mechanical force/torque
- Thus rotor conductors experiences a force/torque, and it tends to rotate the rotor in the same direction of rotating magnetic field (Lenz's Law)
- i.e, the direction of rotor current will be such as to oppose the cause producing it .
- i.e, the relative speed between rotating magnetic field and rotor conductors
- Hence to reduce the relative speed, the rotor starts rotating in the same direction of stator field and tries to attain the same speed of rotating magnetic field

- If rotor runs at the synchronous speed, which is the same speed of the rotating magnetic field, then the rotor will appear stationary to the rotating magnetic field and the rotating magnetic field will not cut the rotor conductors. So, no induced current will flow in the rotor and net torque generated is zero. Hence the rotor speed will fall below the synchronous speed due to load torque and inertia of rotor
- When the speed falls, the rotor windings will cut the rotating magnetic field and a torque is produced and rotor again accelerates
- Due to inertia of rotor, motor will run at a speed which is less than synchronous speed
- The difference between the actual motor speed and the synchronous speed is called the *Slip speed (N_{slip})* = $N_s - N$
- Slip (S) is expressed as the % of synchronous speed.
- Slip is given by $S = \frac{N_s - N}{N_s} * 100\%$

Equivalent circuit of an Induction motor



Equivalent circuit of Rotor



Rotor current frequency, f_r = slip * stator supply frequency

Rotor impedance/phase, $Z_2 = \sqrt{R_2^2 + (sX_2)^2}$

Rotor power factor, $\cos\phi_2 = \dfrac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$

Rotor current/phase, $I_2 = \dfrac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \dfrac{E_2}{\sqrt{\left(\dfrac{R_2}{s}\right)^2 + X_2^2}}$

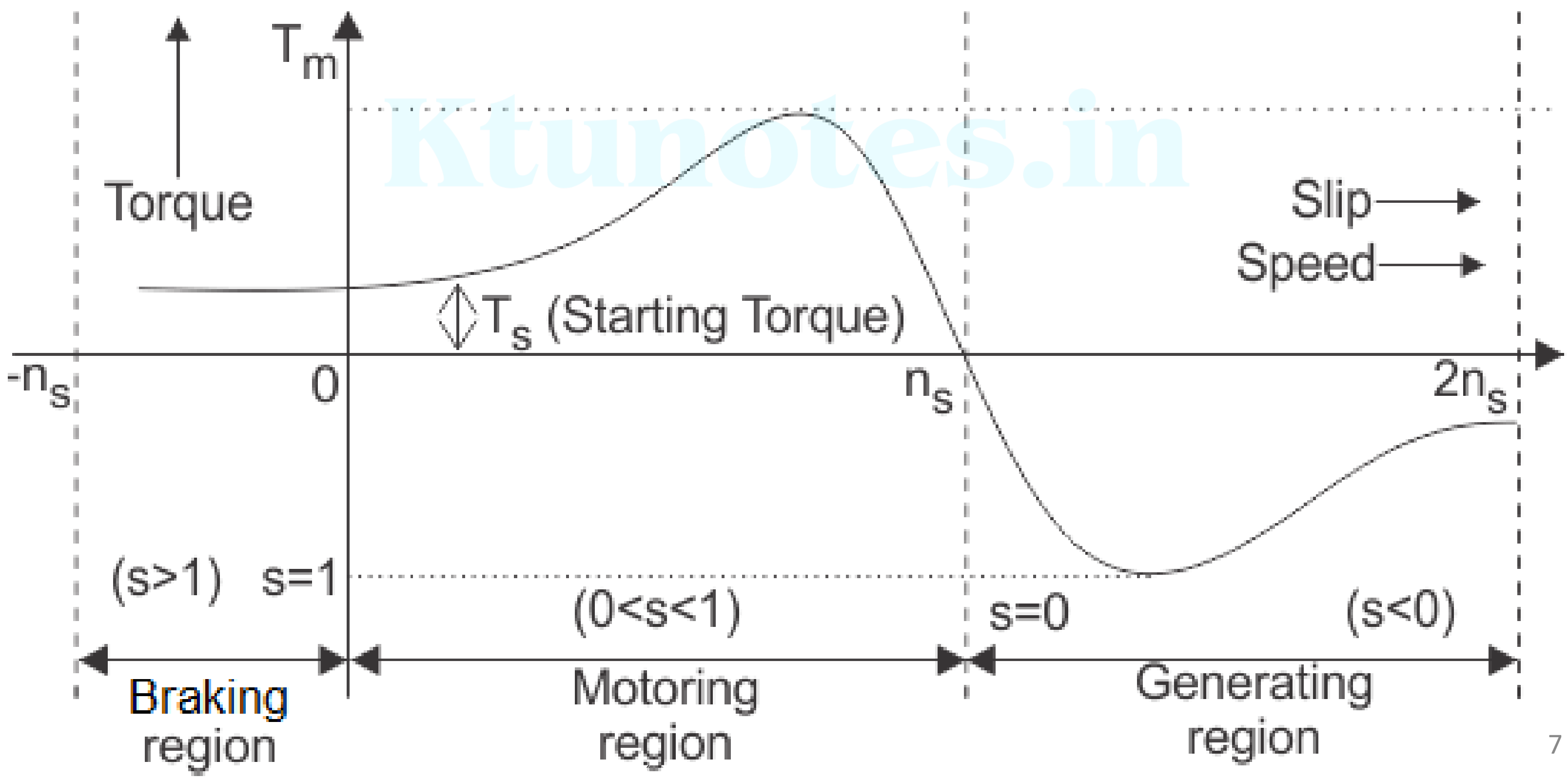
- The rotor resistance (R_2/s) is a function of slip & can be splitted into two parts R_2 & $R_2((1-s)/s)$
- Where R_2 is the rotor resistance & $R_2((1-s)/s)$ is the fictitious resistance, represents the electrical load on rotor
- The power consumed by fictitious resistance ($I_2^2 R_2((1-s)/s)$) represents the mechanical power developed in rotor

- Torque developed in the motor, $T \alpha \phi I_2 Cos\phi_2$

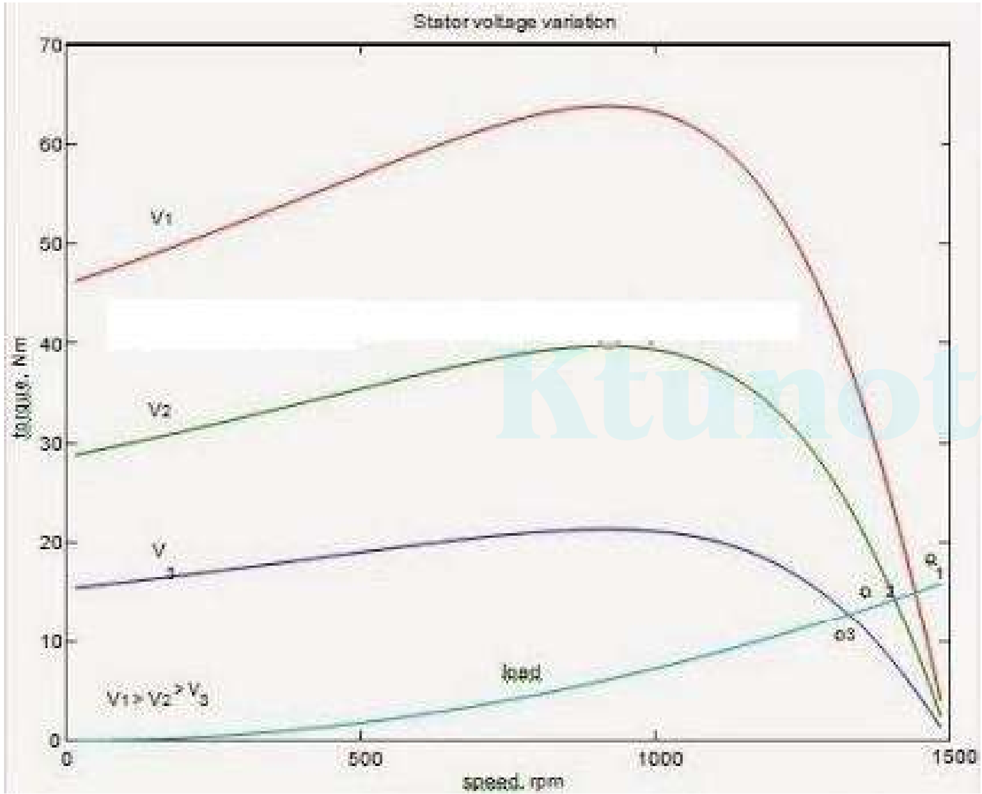
  i.e, $T = KE_2I_2Cos\,\phi_2$ (flux is proportional to voltage)

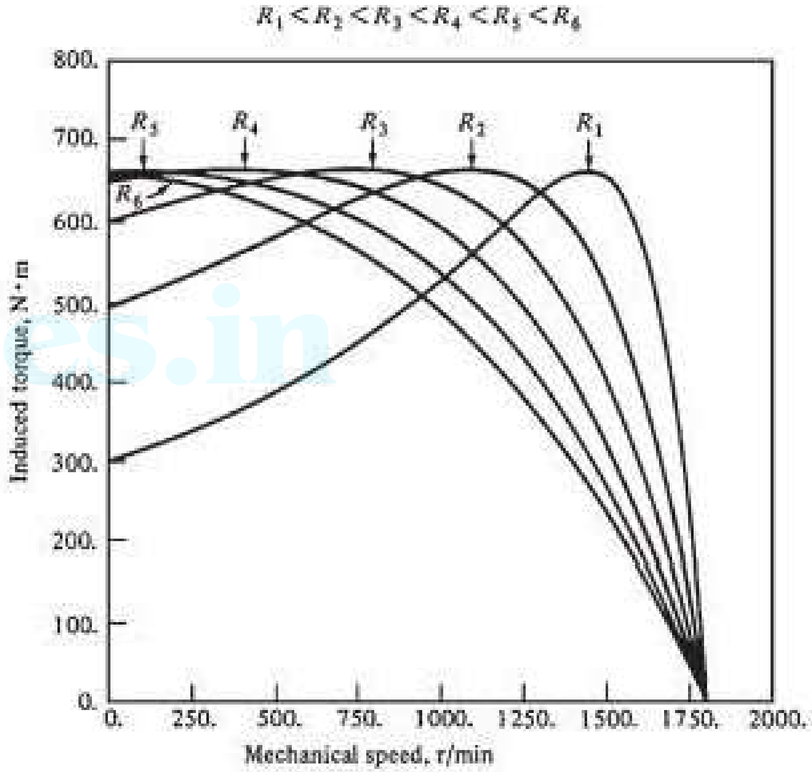
  i.e, $T = \dfrac{KsR_2E_2^2}{R_2^2 + (sX_2)^2}$ (substituting for I_2 & $Cos\phi_2$)

<u>Speed – torque characteristics</u>



- From torque equation,
  - At full load, slip is very low, $(sX_2)^2$ can be neglected & $T \alpha sE_2^2$
  - Also torque can be varied by varying the rotor resistance





Speed control of Induction motor

- Different methods employed for speed control of Induction motor are
  1. Pole changing
  2. Stator voltage control
  3. Supply frequency control
  4. Rotor resistance control
  5. Slip power recovery
- Methods 1, 2 & 3 are applicable for both SQIM & SRIM
- Methods 4 & 5 are applicable only to SRIM

1. Pole changing

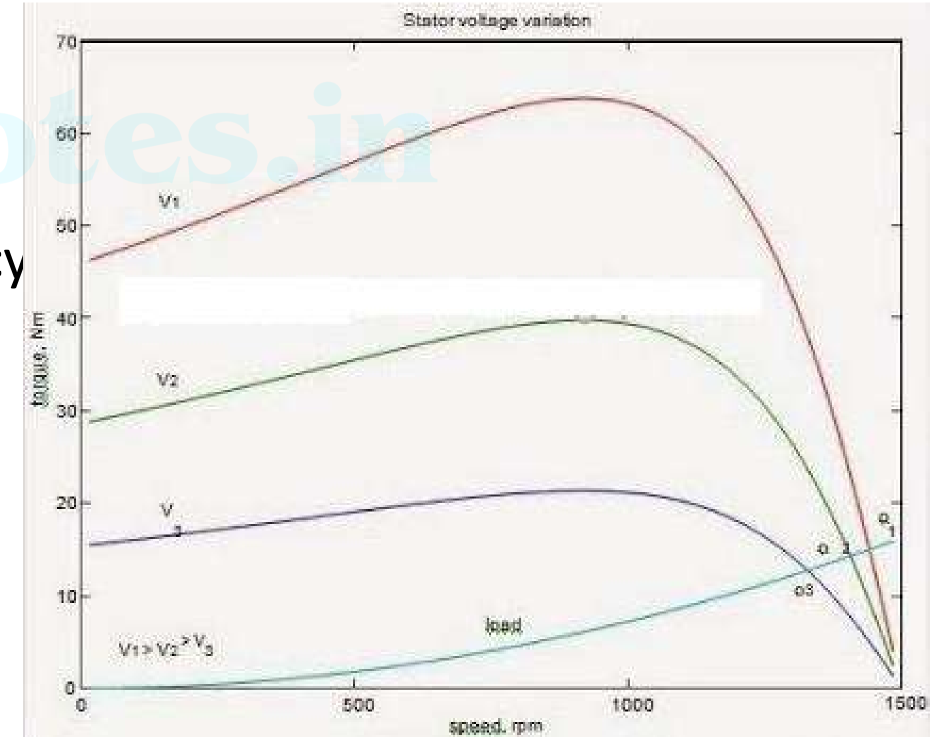
- In an Induction motor, synchronous speed, $N_s=(120f)/P$
- For a given frequency, synchronous speed is inversely proportional to number of poles
- So by changing the number of poles, synchronous speed & therefore speed of motor can be changed

Number of poles	Synchronous speed for frequency of 50Hz
2	3000 rpm
4	1500 rpm
6	1000 rpm
8	750 rpm
10	600 rpm

- Provision for changing the number of poles has to be incorporated in the machine at the manufacturing stage
- Such machines are called pole changing motors or multi speed motors

2. Stator voltage control

- By reducing stator voltage, the speed of an induction motor can be reduced by an amount which is sufficient for speed control of some fan & pump applications
- From torque equation, Torque α $Voltage^2$
- So when voltage is reduced to reduce speed, for the same current motor develops lower torque which is suitable for some pump & fan applications
- Motor efficiency is given by,
  $\eta = P_m/P_g = (1-s)$
- This equation shows that efficiency falls with decrease in speed
- Speed control is essentially obtained by dissipating a portion of rotor input power in rotor resistance & this over heats rotor



- Variable voltage for speed control is obtained by using ***AC voltage controller***
- By stator voltage control, speed control below base speed is only possible

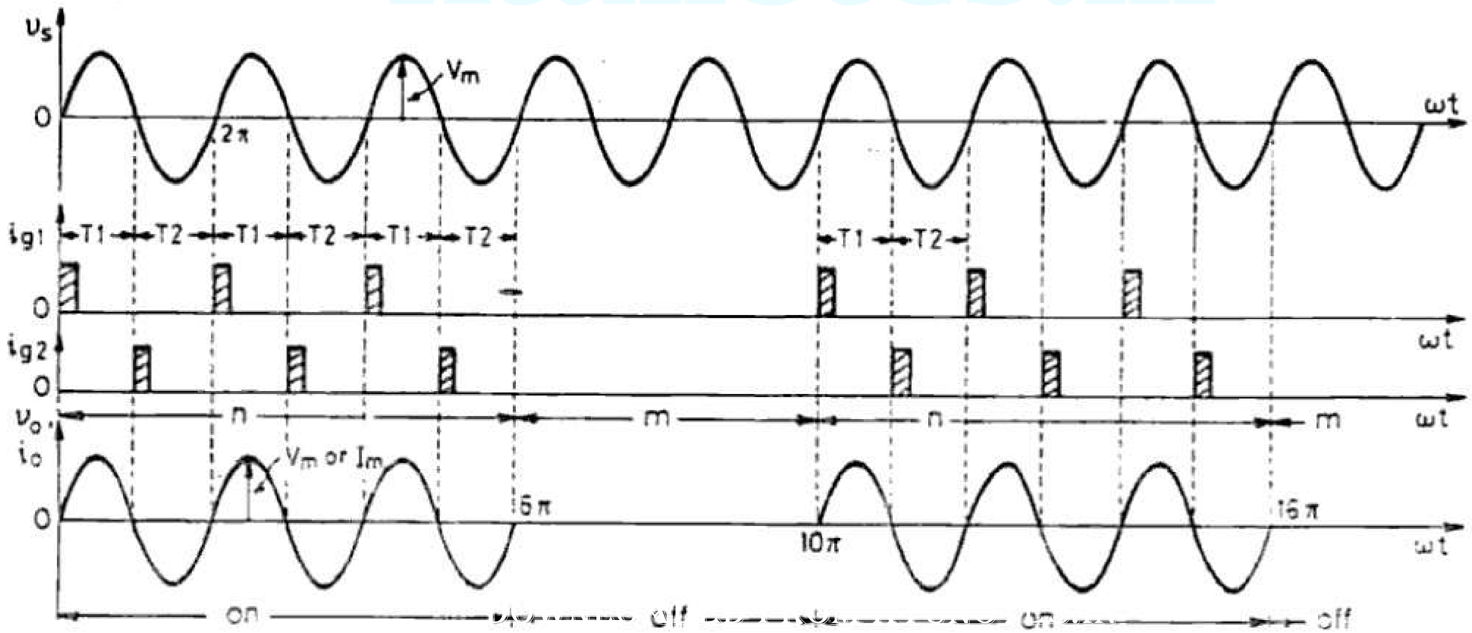
Stator voltage control by using AC voltage controller

- AC voltage controller produces a variable voltage AC voltage of same frequency from a fixed AC voltage
- Varies only output voltage magnitude but not the frequency
- The speed of fans & pumps are commonly controlled by a single phase or 3 phase AC voltage controller
- Domestic fans & pumps are always single phase & we use a single phase AC voltage controller to control speed
- Industrial fans & pumps are usually 3 phase & we use a 3 phase AC voltage controller
- AC voltage controller use either TRIAC (for low power rating motors) or anti parallel thyristors.
- Speed control is obtained by varying the firing angle

- Since AC voltage controller allow a step less control of voltage from its zero value, they are used for soft starting also
- In AC voltage controller two control strategies are possible

1. Integral cycle control or ON-OFF control

- Here thyristors are employed as switches to connect the load circuit to the source for a few cycles of source voltage and then disconnect it for another few cycles
- In industry, integral cycle control is used in applications in which mechanical or thermal time constant is high



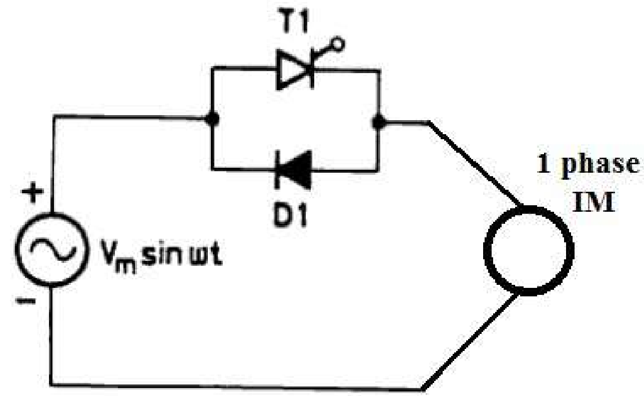
2. Phase control

- Here the power flow from source to the load is controlled by varying the firing angle of the thyristor/TRIAC in each cycle
- Phase angle control induces more harmonics in to the supply system than ON-OFF control
- Different configurations of phase controlled AC voltage controllers are

• 1 phase half wave AC voltage controller
• 1 phase full wave AC voltage controller
• 3 phase half wave AC voltage controller
• 3 phase full wave AC voltage controller

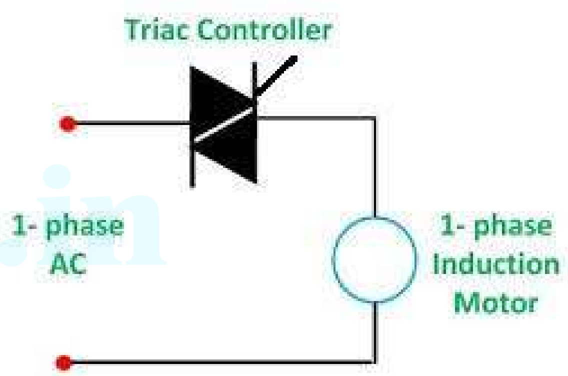
<u>1 phase half wave AC voltage controller</u>

- It consist of one thyristor and an anti parallel diode
- Voltage applied to motor stator is varied by varying the firing angle of thyristor



<u>1 phase full wave AC voltage controller</u>

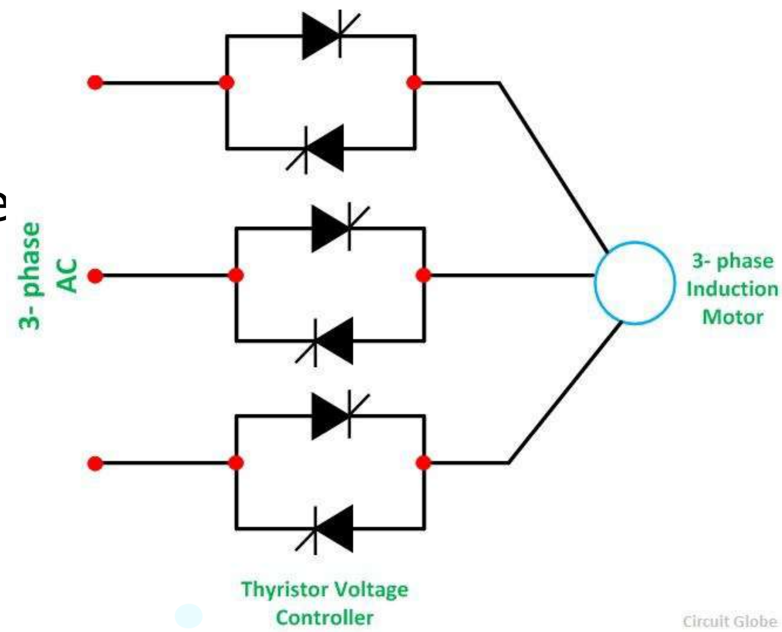
- Used for the speed control of 1 phase fans/pumps
- Uses anti parallel thyristors/TRIAC
- 1 quadrant drive

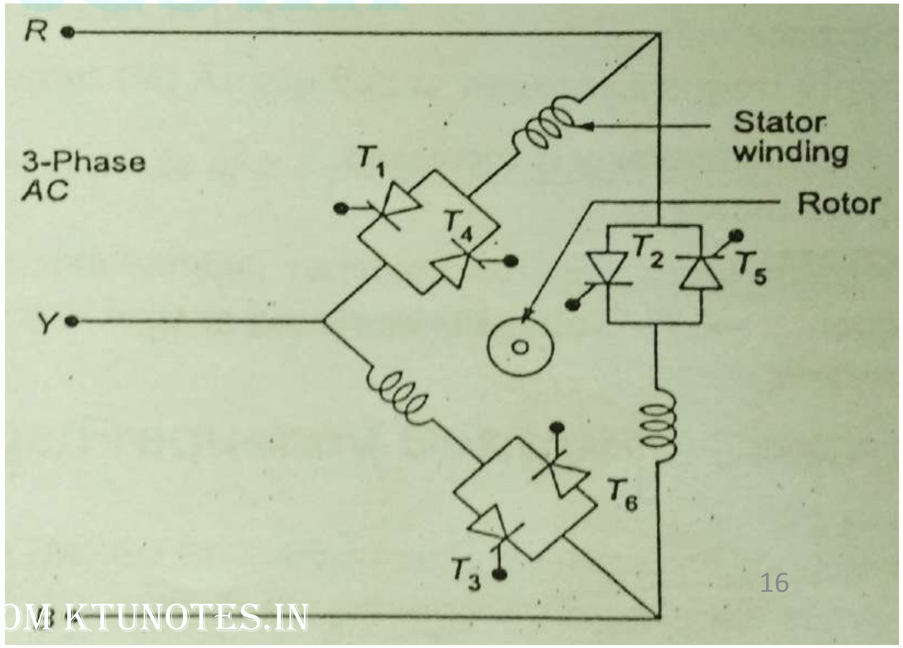


<u>3 phase full wave AC voltage regulator for star connected load</u>

- Usually anti parallel thyristors are used for large power rating motors
- 1 quadrant drive
- Each thyristor pair carries line current

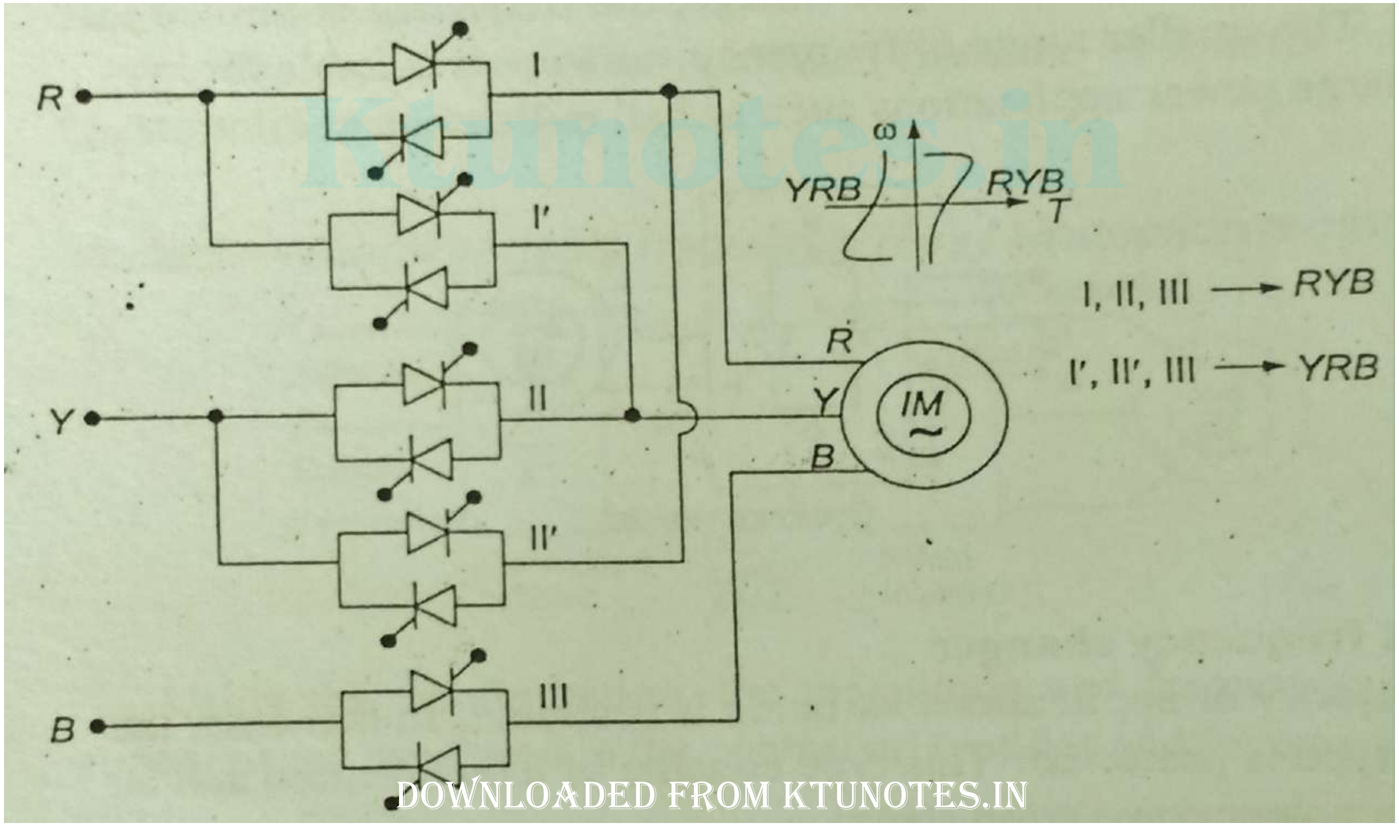
- This connection can be used for both star & delta connected stator windings
- In delta connection, line current is more & we need to use thyristors of large current rating
- So for *delta connection* we use another configuration as shown in 2nd figure
- Here each thyristor pair carries the phase current
- The thyristor current rating is reduced by a factor of √3





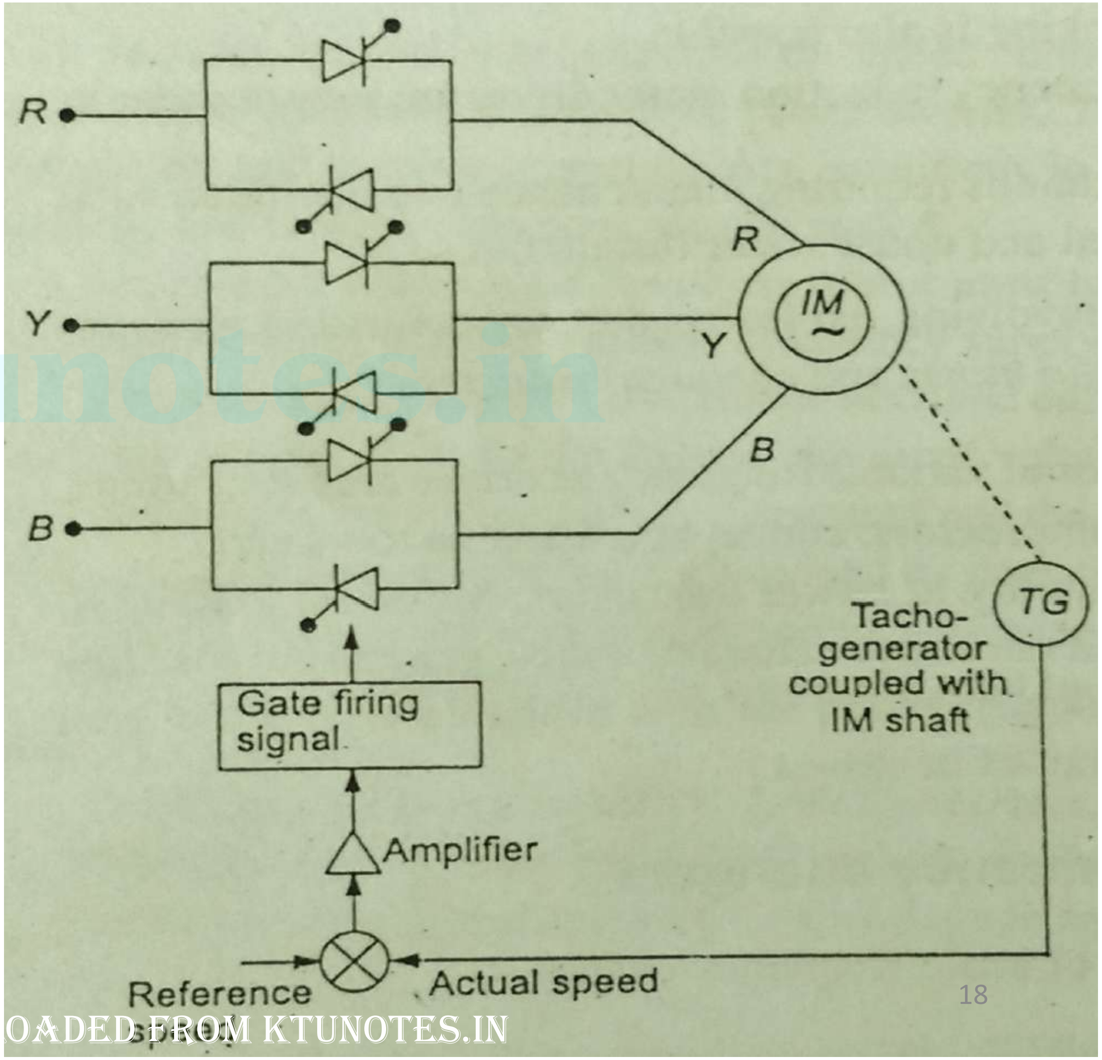
4 quadrant Induction motor drive using AC voltage controller

- It contain 5 thyristor pairs
- Thyristor pairs I, II & III are operated to get operation in quadrant I & IV
- Thyristor pairs I’, II’ & III are operated to get operation in quadrant II & III
- Here regenerative braking is not possible, plugging is possible



Closed loop control of Induction motor using AC voltage controller

- The block diagram of closed loop control of 3 phase induction motor using AC voltage controller is shown in figure
- It consist of 3 phase induction motor, power circuit, tacho generator & control circuit
- Tacho generator generates a voltage proportional to actual motor speed
- Actual speed is compared with reference speed



- The difference in speed is compared , error is amplified & fed to firing control circuit
- The firing angle changes according to speed error
- When firing angle changes, motor terminal voltage changes & hence speed of motor

Advantages of stator voltage control

- Simple circuit
- Compact & less weight
- Quick response
- Economical
- Provides step less control of voltage
- Can be used for soft starting of motors

Drawbacks

- Input power factor is low
- Voltage & current waveforms are distorted due to harmonics

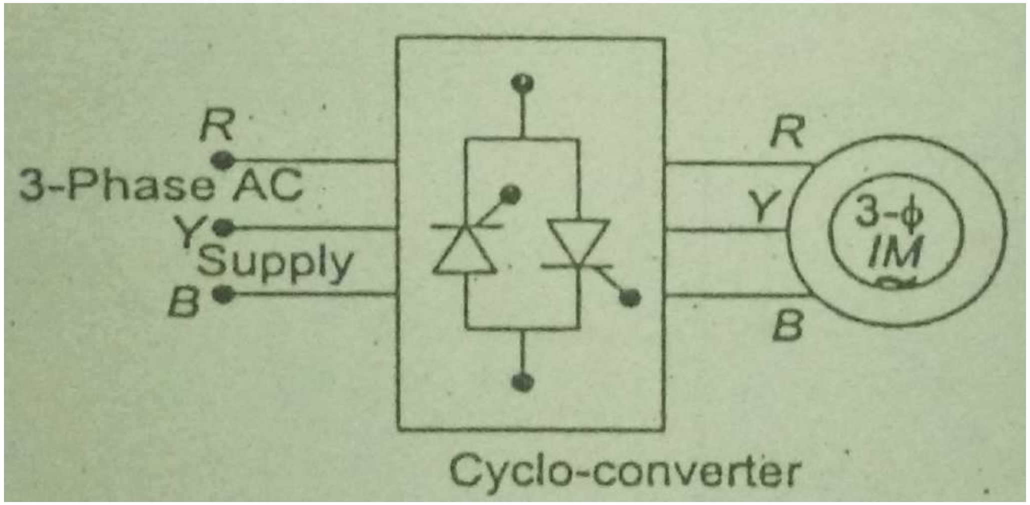
- Maximum torque decreases because stator voltage decreases
- At low speeds motor current is high & arrangements needed to limit the current
- Regeneration is not possible

3. Supply frequency control

- In an induction motor, synchronous speed & there by the motor speed can be controlled by varying the supply frequency
- i.e, $N_s = (120f)/P$, $N_s \alpha f$
- The supply frequency is constant. So to change frequency we need a frequency changer circuit
- Commonly used frequency changer circuits are
  - Cycloconverters
  - Inverters

<u>Cycloconverters</u>

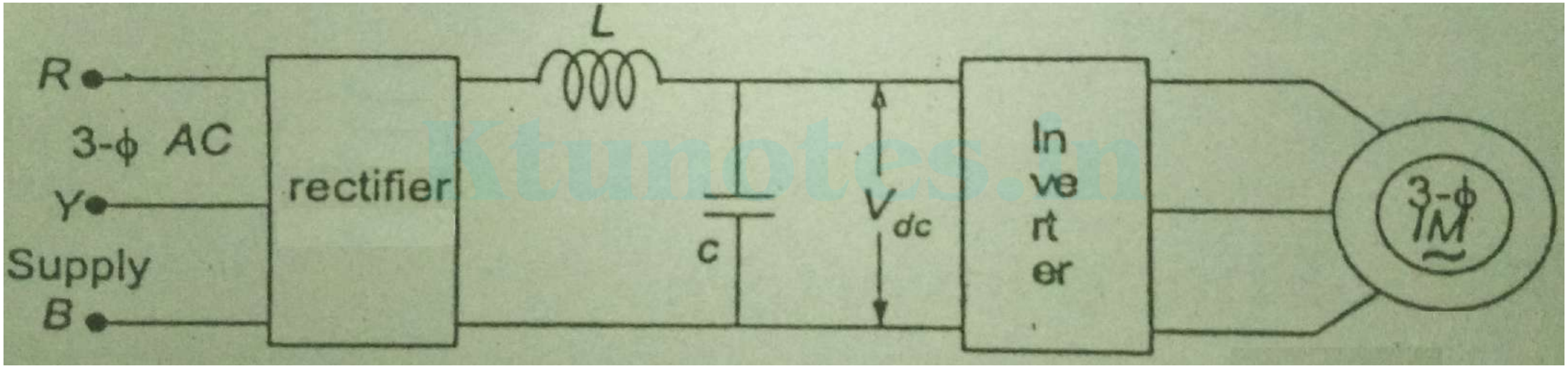
- Converts AC supply frequency to a variable frequency
- When operating at low frequencies, output has low harmonic content



- Harmonic content increases with frequency making it necessary to limit the output frequency to 40% of source frequency
- Maximum speed is restricted to 40% of synchronous speed
- This will provide a smaller range of frequency variation, which is suitable for low speed, large power applications like ball mills, cement kiln etc
- The drive has regenerative braking capability
- 4 quadrant operation is obtained by reversing the phase sequence of motor terminal voltage

Inverters

- It include rectification & inversion of supply
- It can be of two type
    - Voltage source inverter
    - Current source inverter

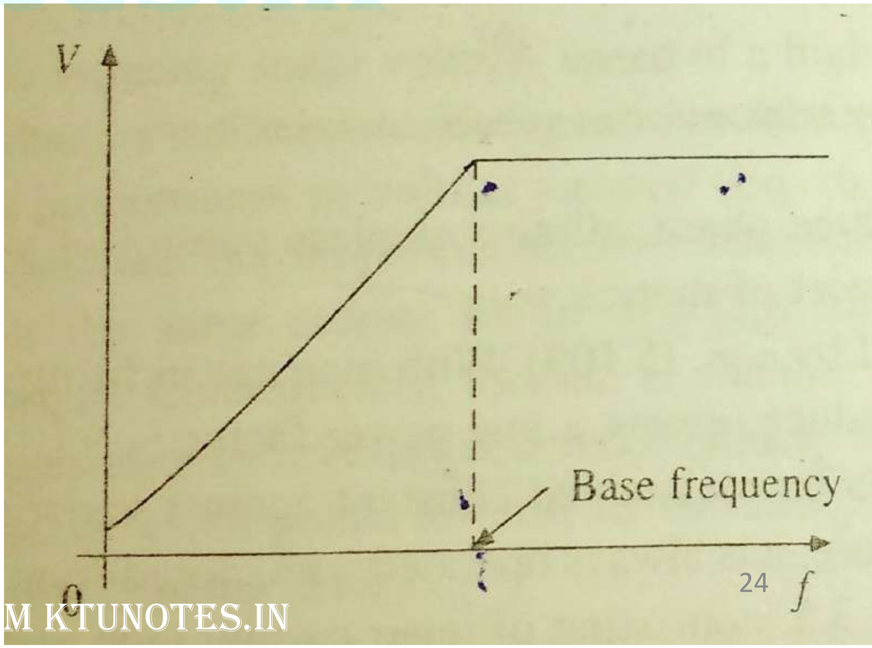


- The rectifier can be controlled/uncontrolled
- Inverter can be PWM/square wave
- Here output voltage & frequency can be controlled

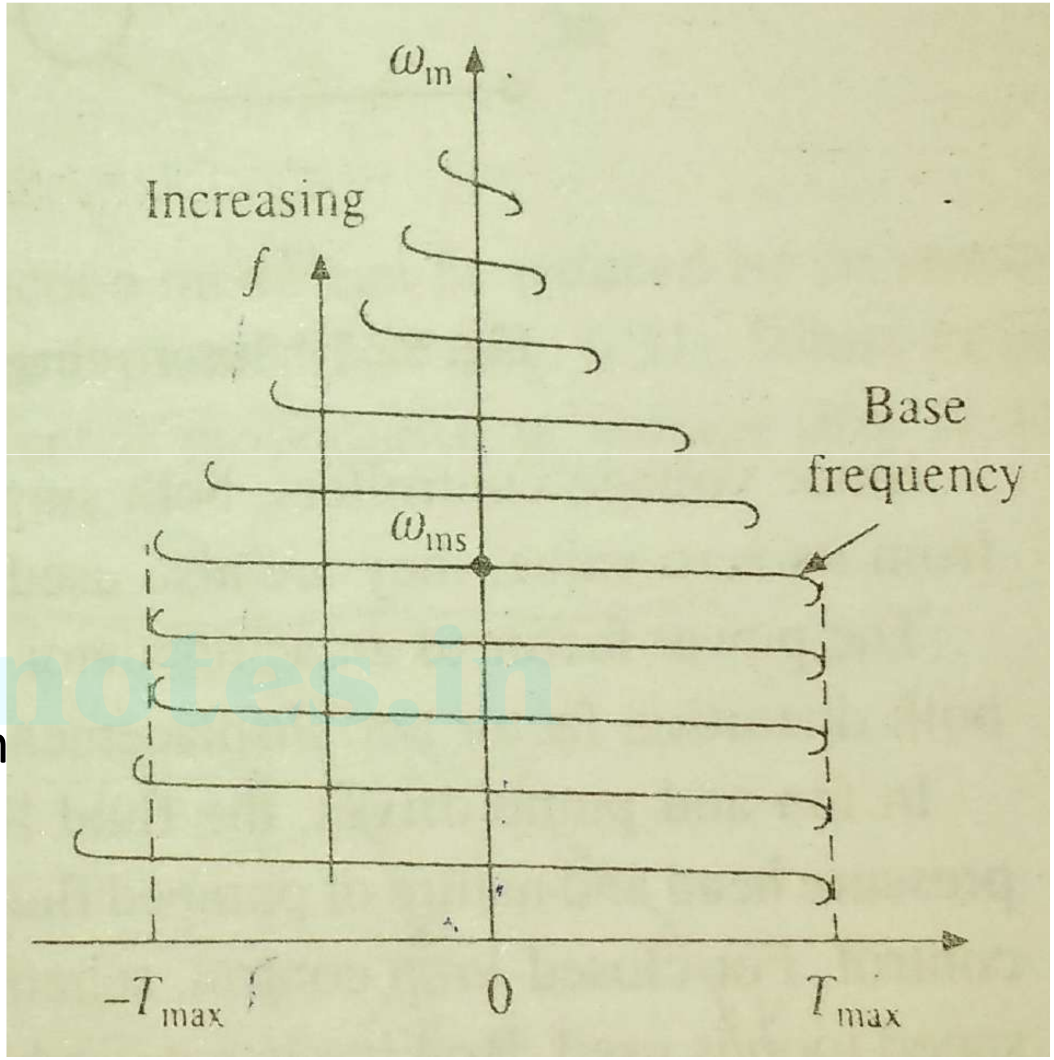
V/f control of Induction motor

- In an Induction motor, synchronous speed & there by motor speed can be controlled by changing the supply frequency
- But when supply frequency is varied, it will affect the performance of the motor
- In an Induction motor, terminal voltage α (frequency x flux) (similar to a transformer)
- An increase in supply frequency, without a change in terminal voltage causes an increase in flux
- Increase in flux will saturate the motor
- As a result, magnetizing current increases, line voltage & current gets distorted, core loss increases & produces a high pitch acoustic noise
- Also when frequency is increased to increase speed by keeping the voltage constant, flux decreases
- As a result, torque produced by motor decreases
- So an increase/decrease in flux beyond rated value is undesirable

- There fore, variable frequency control below rated speed is carried out by varying the terminal voltage along with frequency so as to maintain the V/f ratio constant at rated value
- For speed above rated speed, V/f ratio cannot be maintained constant since terminal voltage cannot be increased above rated value
- So terminal voltage is kept at rated value and frequency is varied to get speed above rated speed
- As frequency increases, flux decreases & torque produced by motor decreases
- The variation of terminal voltage with frequency in v/f control is shown in figure

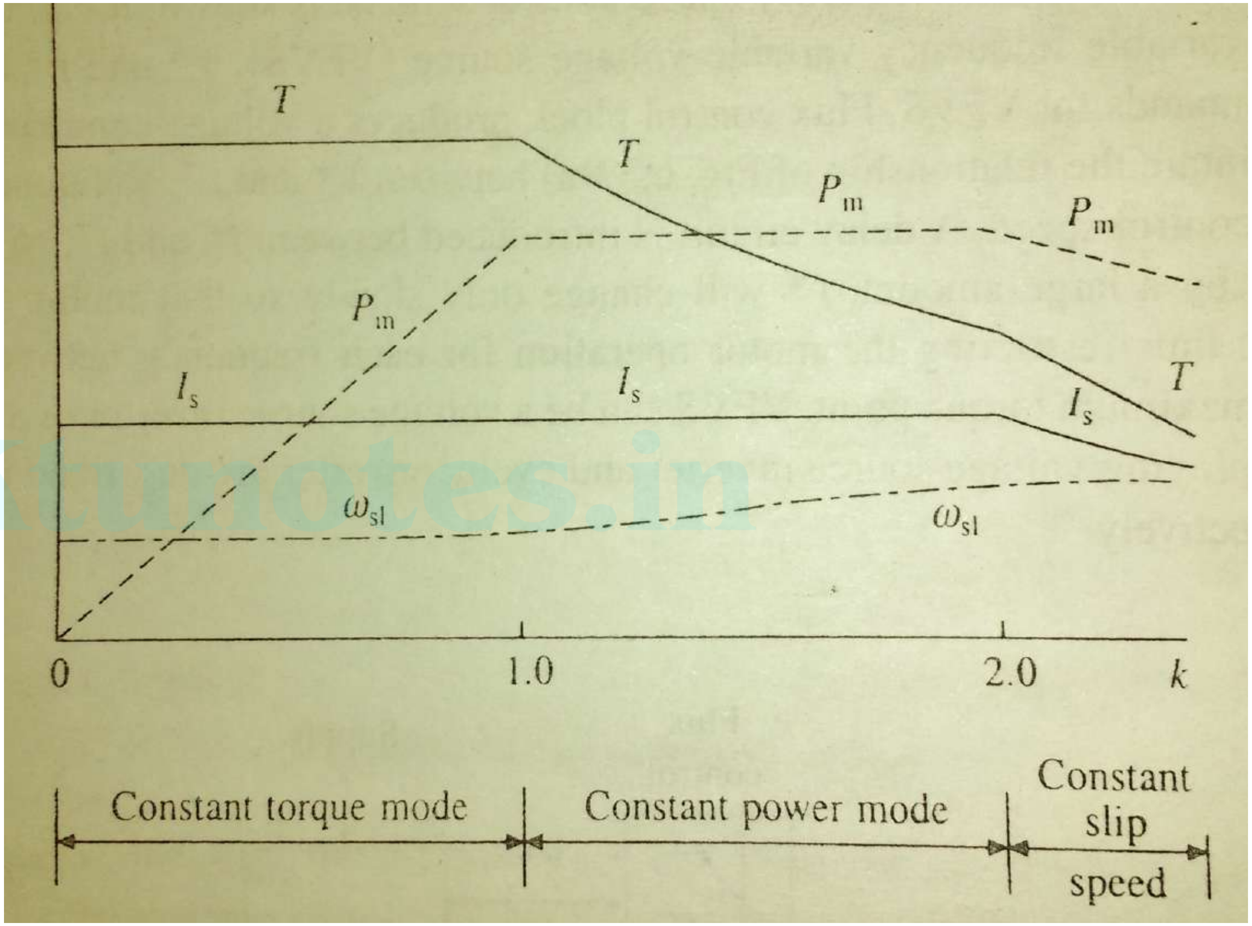


- The variation of speed-torque characteristics in V/f control is shown in figure
- Variable frequency control provides good running & transient performance due to following reasons
  - Speed control & braking operation are available from zero speed to above rated speed
  - Losses are low, efficiency & power factor are high
  - Drop in speed from no load to full load is small



Constant torque & Constant power operation

- The variation of maximum torque & power with frequency is shown in figure below
- From zero to base speed, the motor operates in constant torque mode
- From base speed to twice the base speed, motor operates in constant power mode
- Beyond this speed, the machine operates at a constant slip speed



- Constant torque & constant power modes are present in the v/f control of induction motor below & above base speed

<u>Constant torque mode</u>

- Induction motor operates in constant torque mode for speed control below base speed by v/f control
- To get speed below base speed, supply frequency is decreased by keeping v/f ratio constant
- The motor current remains constant at maximum rated value
- So as voltage is varied to keep v/f ratio constant, power produced by motor varies
- Since v/f ratio & hence flux is constant in machine, torque produced by machine remains constant & we get constant torque operation

<u>Constant power mode</u>

- Induction motor operates in constant power mode for speed above base speed by v/f control
- To get speed above base speed, supply frequency is increased
- Stator supply voltage cannot be increased above rated voltage, so v/f ratio cannot be kept constant
- The motor current remains at maximum rated value

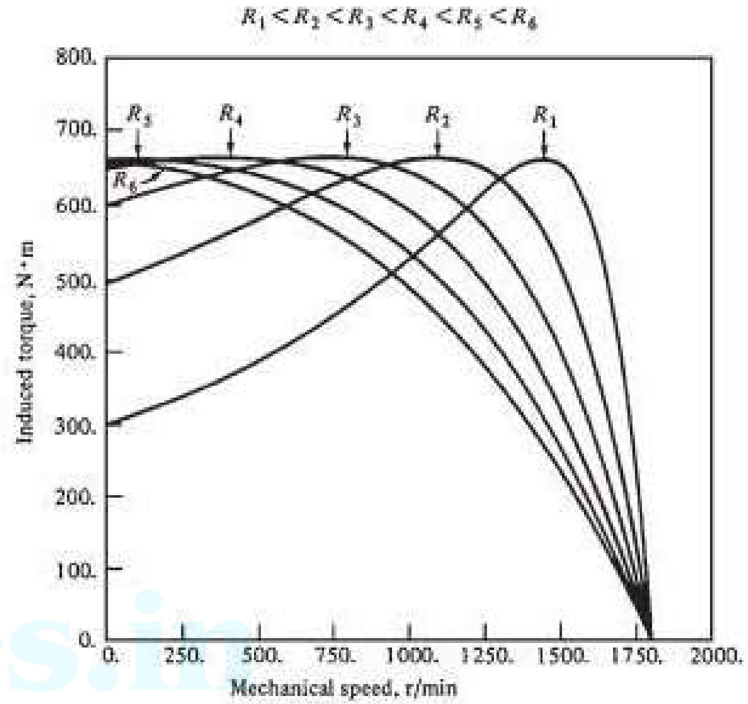
- Since motor current & voltage are constant, power produced by motor remains constant & we get constant power operation
- When frequency increases, flux in the machine decreases & torque produced by machine also decreases as shown
- So constant power mode is also called field weakening mode of an induction motor
- For speed above twice base speed, machine operates at a constant slip speed & maximum current & power decreases
- The operation in this region is used in drives requiring wide speed range but low torque at high speeds, like in traction
- This region of operation is also called high speed series motoring region

- The variable voltage variable frequency supply for induction motor control can be obtained from an *inverter* or *cycloconverter*

Rotor resistance control

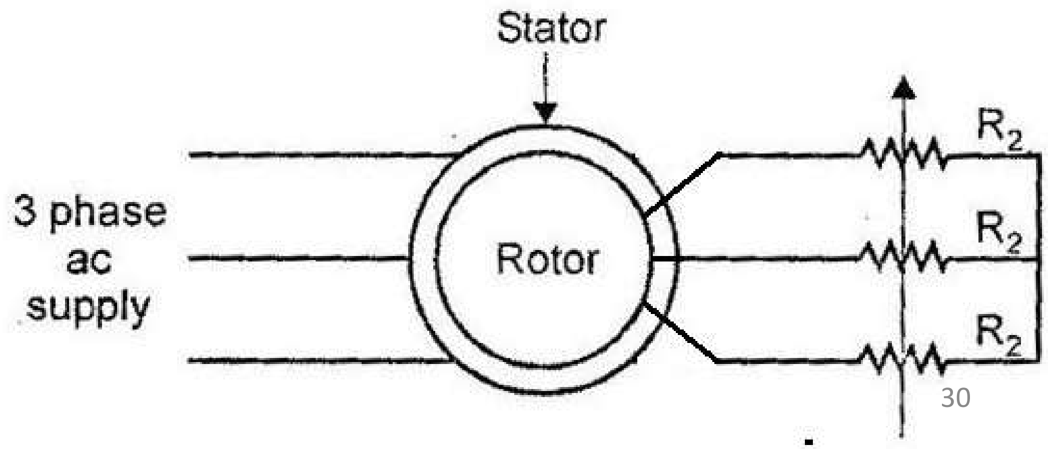
- Speed of an induction motor can be controlled below rated speed by rotor resistance control method
- This method is applicable only to SRIM
- Torque equation of an induction motor is, $T \alpha \frac{sR_2 E_2^2}{R_2^2 + (sX_2)^2}$
- When machine runs at near synchronous speed, slip is very low & the term $(sX_2)^2$ become very small compared to R_2
- So it can be neglected, i.e, T α (s/R_2)
- If the load torque (T) remains constant, when R_2 varies 's' also varies
- So slip & hence speed can be controlled by varying rotor resistance
- The additional rotor resistance improves starting torque
- But it causes additional wastage of power in rotor resistance in the form of I^2R loss

- The variation of torque – speed charac- -teristics with variation in rotor resistance is shown in figure



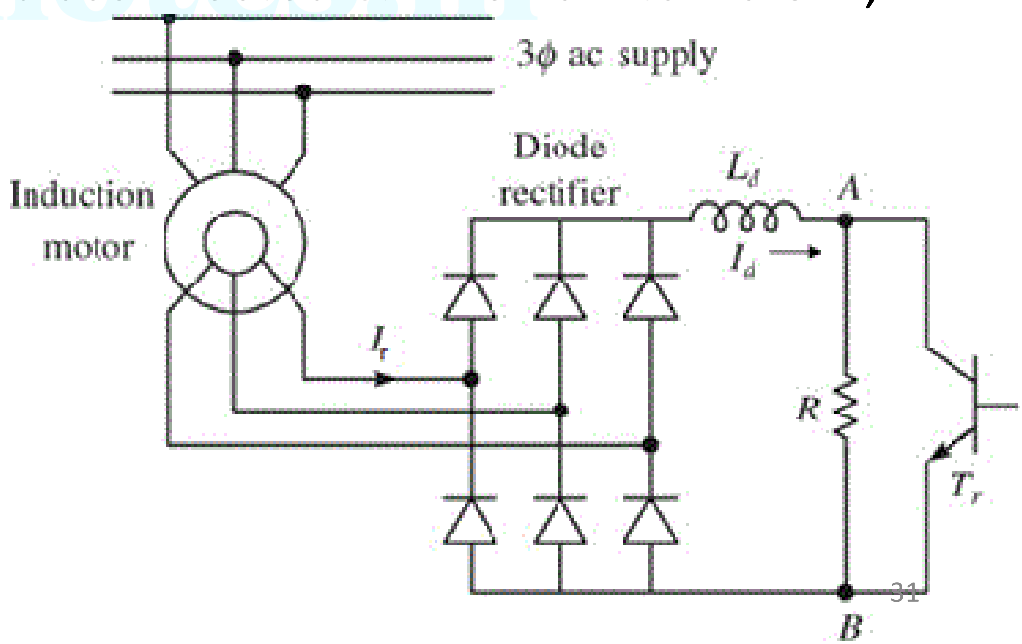
<u>Conventional Rotor resistance control</u>

- Here external resistance is connected to rotor circuit as shown & is varied mechanically to get desired performance



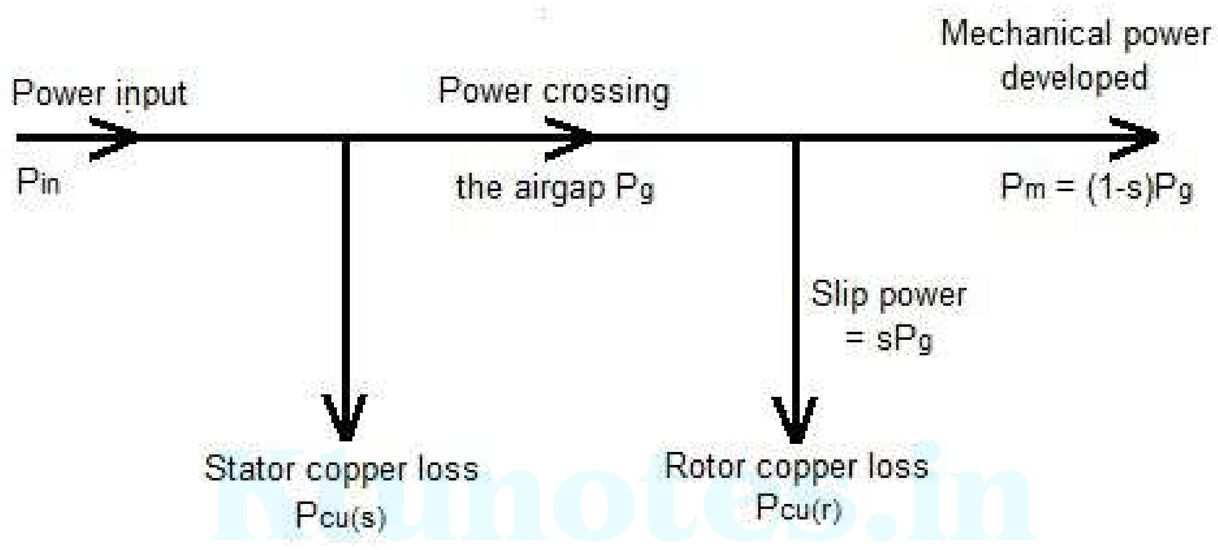
Static rotor resistance control/Rotor chopper speed control

- Here instead of mechanically varying the rotor resistance, it is varied statically by using the principle of a chopper
- The AC output voltage of rotor is rectified by a diode bridge rectifier & fed to a parallel combination of fixed resistance R and a self commutated switch T_r
- When the switch operates, the resistance R periodically connects & disconnects to DC supply
- When switch is ON, resistor is disconnected & when switch is OFF, resistor is connected
- So by controlling the ON time of switch (duty ratio), the effective rotor resistance can be varied



Principle of Slip power recovery scheme

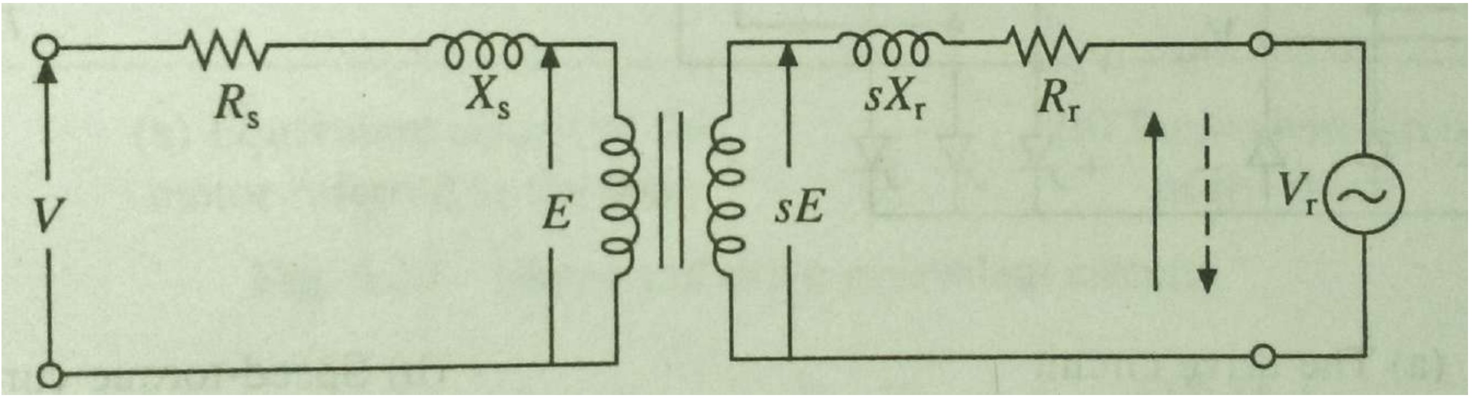
- The power flow diagram of an induction motor is shown in figure



- P_{in} is the total stator input power
- Some power is wasted in stator winding as stator copper loss ($P_{cu(s)}$)
- Remaining power is crossing the air gap to reach rotor. i.e air gap power $P_g = P_{in} - P_{cu(s)}$
- A portion of P_g is wasted in rotor resistance as rotor copper loss. i.e given by $P_{cu(r)} = sP_g$

- Remaining air gap power is converted into mechanical power, $P_m = (1-s)P_g$
- The portion of air gap power which is not converted into mechanical power is called ***slip power***
- Normally in an induction motor, slip power is wasted in rotor resistance as heat
- When rotor resistance control is used for speed control of SRIM, we are controlling the slip power to control the speed of motor
- When rotor resistance increases, speed of motor decreases
- Actually when rotor resistance increases, rotor copper loss increases. i.e, slip power increases. As a result mechanical power developed decreases because, $P_g = P_{cu(r)} + P_m$
- Instead of wasting the slip power in rotor resistance, we can recover this power & can be fed back to supply
- This method of speed control is called ***slip power recovery scheme***

- The equivalent circuit of a SRIM is shown in figure



- Here, the external rotor resistance which consume the slip power is represented as a voltage source V_r
- Now we can write, $P_m = P_g – P_r$, where P_r = power absorbed by the source V_r
- The magnitude & sign of P_r can be controlled by controlling the magnitude & phase of V_r
- When $P_r = 0$, motor runs on its natural speed torque characteristics
- A positive P_r will reduce P_m & there fore, motor speed decreases for same torque
- When $P_r = P_g$, then P_m & speed = 0

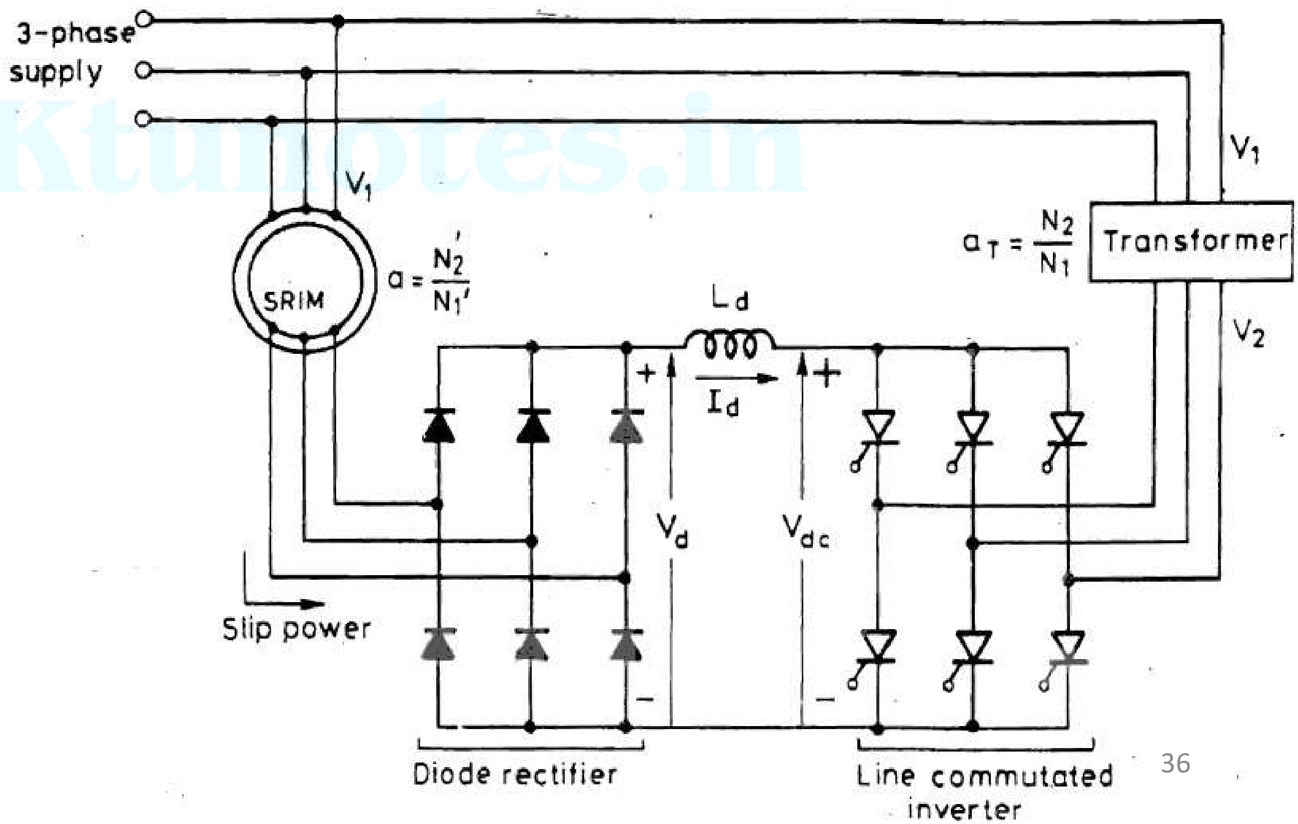
- So variation of P_r from 0 to P_g will allow speed control from synchronous to zero speed
- Polarity of V_r for this operation is shown by continuous line
- When P_r is negative, i.e, V_r act as a source of power, P_m will be larger than P_g
- Motor will run at a speed higher than synchronous speed
- Polarity of V_r for speed control above synchronous speed is shown by dotted line
- Here speed control below synchronous speed is obtained by controlling the slip power, the same approach used in rotor resistance speed control
- For speed above synchronous speed, additional power to be supplied to rotor
- These methods of speed control are called slip power recovery schemes
- There are two slip power recovery schemes, they are

  1. Static Kramer Drive
  2. Static Scherbius Drive

1. Static Kramer Drive

- The circuit configuration is shown in figure
- Here the slip frequency power from rotor is converted to DC voltage, which is then converted to line frequency AC and pumped back to the AC source
- Here slip power can flow only in one direction, so this drive offers speed control below synchronous speed only

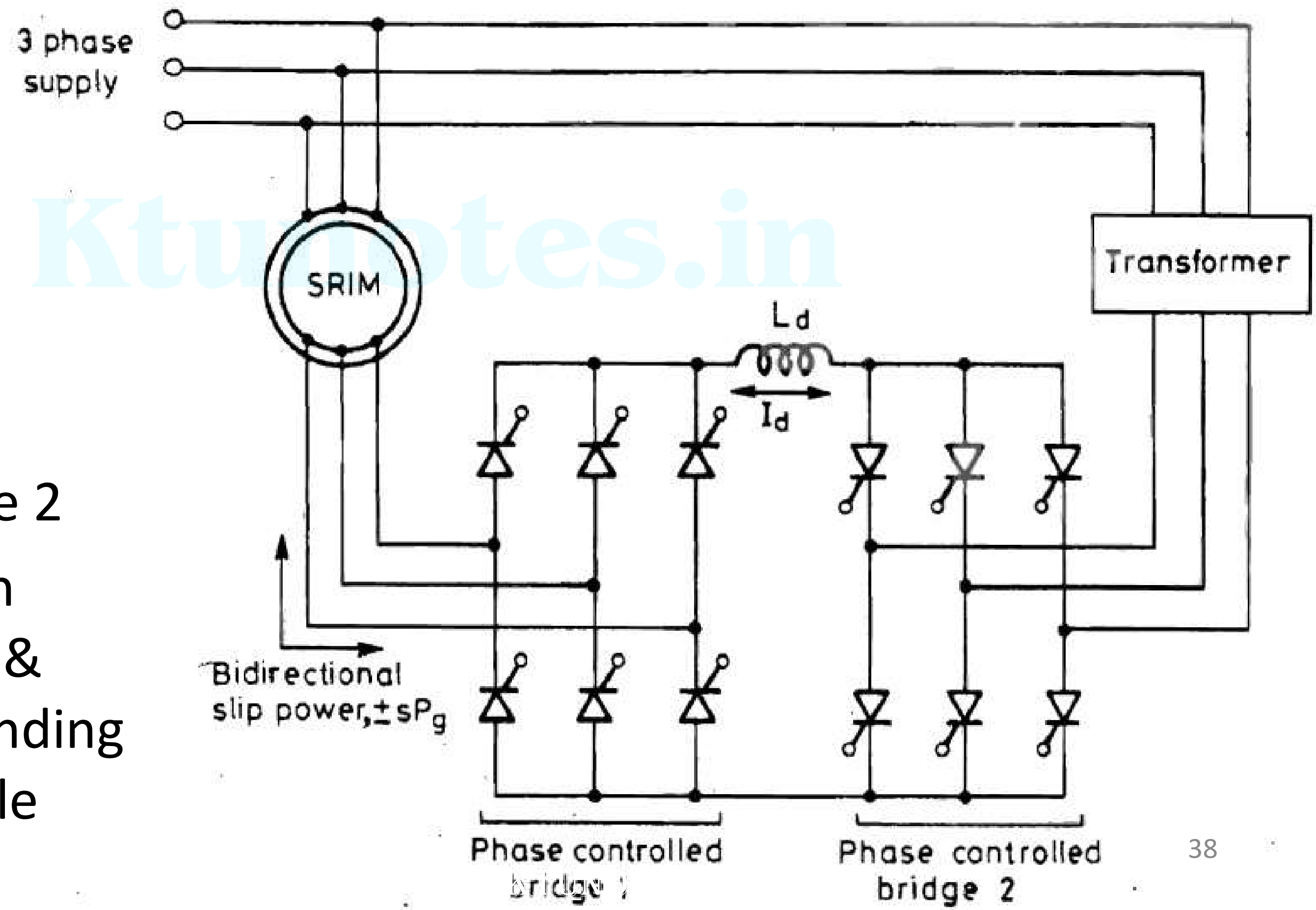
- The drive input power is the difference between motor input power & power fed back



- The slip power from rotor is rectified to DC voltage by a diode bridge
- Inductor L_d smoothens ripples in rectified voltage V_d
- The DC voltage is then converted to AC voltage at line frequency by a line commutated inverter & fed back to AC supply
- In practice the rotor circuit voltage is less than stator supply voltage, so a 3 phase transformer is often required between AC supply & the inverter
- The slip power fed back to the source can be controlled by controlling the firing angle of inverter
- On analysis we can see, slip is directly proportional to $\cos\alpha$
- Maximum value of α is restricted to 165° for safe commutation of inverter thyristors
- So slip can be controlled by varying α from 90 to 165° & hence speed of motor
- This drive got applications in fan & pump drives which require speed control below synchronous speed in small range only
- Here instead of wasting slip power in rotor resistance, it is fed back to source. So drive has a high efficiency

2. Static Scherbius Drive

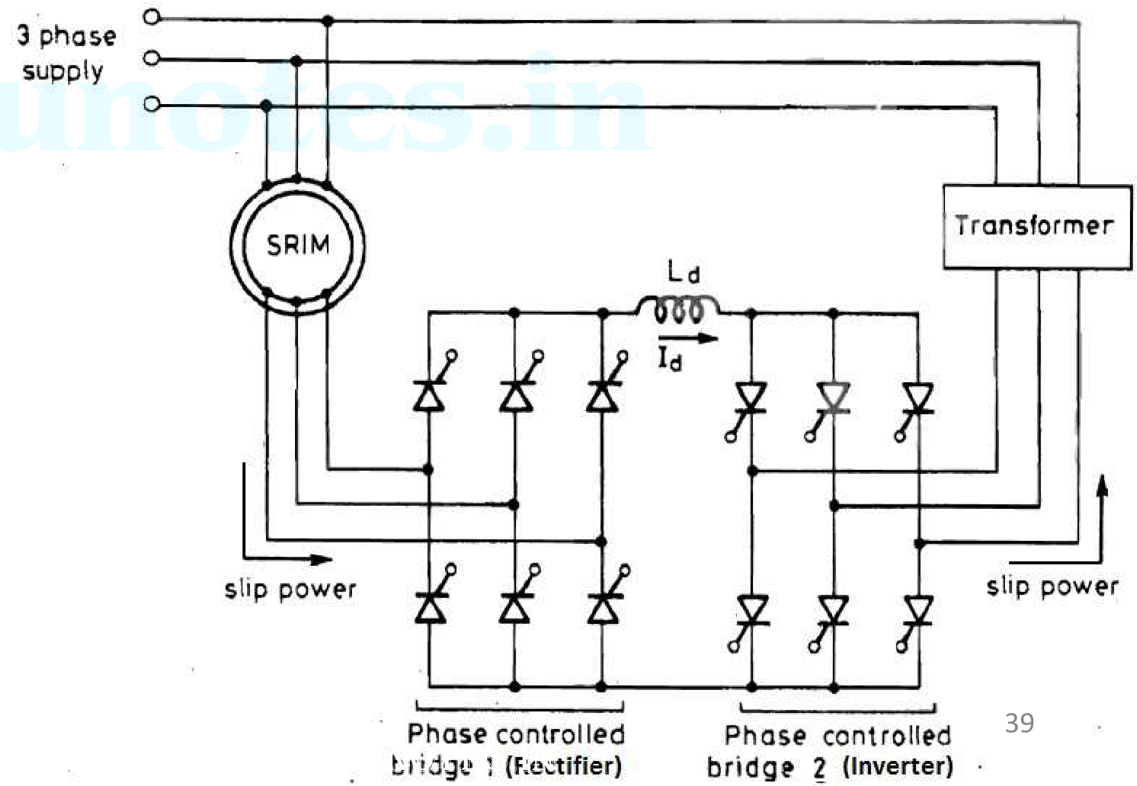
- In static Kramer Drive, speed of SRIM can be varied only below synchronous speed
- For speed control of SRIM above & below synchronous speed, static Scherbius Drive is used
- The drive is shown in figure
- It consist of two fully controlled bridge circuits - Bridge 1 & Bridge 2
- Both bridges can work as rectifier & as inverter depending on the firing angle



- By using this drive, speed control above & below synchronous speed is possible
- So it got two modes of operation

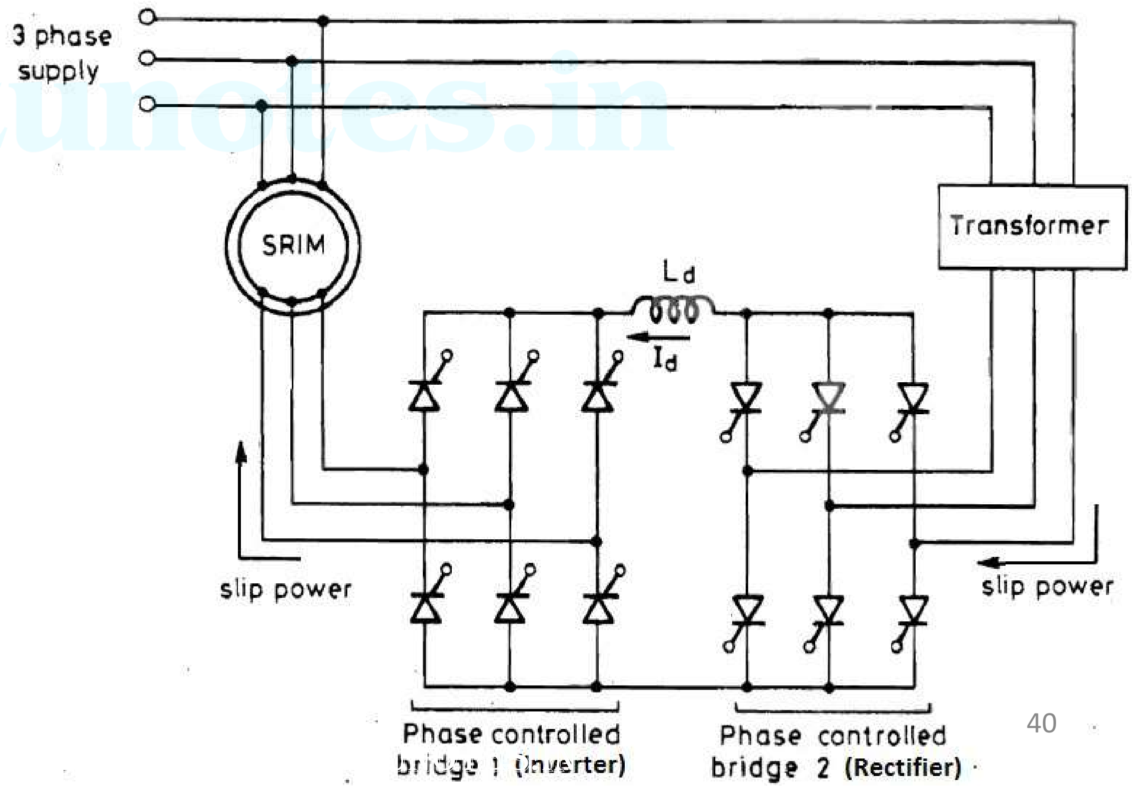
Mode 1 – Sub synchronous speed operation

- In this mode of operation, slip power is removed from the rotor circuit & is pumped back into the AC supply
- Now, Bridge 1 works as rectifier (firing angle less than 90) & Bridge 2 works as inverter (firing angle greater than 90)
- The slip power flows from rotor circuit to bridge 1, DC link, bridge 2, transformer and returned to supply



Mode 2 – Super synchronous speed operation

- For super synchronous speed control, additional power is fed into the rotor circuit at slip frequency
- For super synchronous speed control, bridge 1 works as an inverter (firing angle greater than 90) & bridge 2 works as a rectifier (firing angle less than 90)
- The power flow is now from supply to transformer, bridge 2, bridge 1 & to rotor circuit



- This drive is expensive than static Kramer drive, because it requires 2 controlled bridges & its control circuitry
- Here when motor speed reaches near synchronous speed, magnitude of rotor induced voltage is not sufficient to provide line commutation of thyristors. So we provide forced commutation.

Cycloconverter Scherbius drive

- Here the bridge converter system is replaced by a 3 phase controlled line commutated cycloconverter
- The slip power flow in both direction